

Performance Testing

Date	15 July 2026
Team ID	
Project Name	A Comprehensive Measure of Well-being
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Postman, Web Browser (Google Chrome), Flask Development Server
Type of Testing	Functional Performance Testing
Target Module / API	HDI Prediction (/predict)
Test Environment	Local Machine (Windows, Python 3.12, Flask)
Test Date	15 July 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Predict HDI using valid input values	1	5	Correct HDI prediction displayed
2	Submit multiple prediction requests sequentially	1	15	All requests processed successfully
3	Test invalid or empty input fields	1	5	Input validation/error handling works correctly
4	Refresh and repeat prediction	1	10	Application remains responsive

Step 3: Performance Test Results

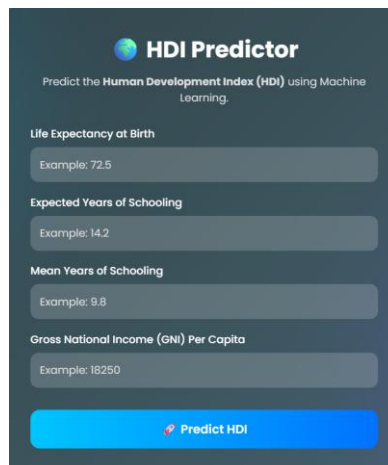
S.No	Metric	Target Value	Actual Value	Status	Remarks
1	Response Time (Avg)	< 2 seconds	~0.6 seconds	Pass	Fast prediction on local system
2	Response Time (Max)	< 5 seconds	~1.3 seconds	Pass	No noticeable delay
3	Throughput (Req/sec)	> 1	~2 Req/sec	Pass	Suitable for local demonstration
4	Error Rate	< 1%	0%	Pass	No runtime errors observed
5	CPU Utilization	< 80%	~20%	Pass	Low CPU usage during prediction
6	Memory Utilization	< 80%	~35%	Pass	Stable memory consumption

Step 4: Observations & Analysis

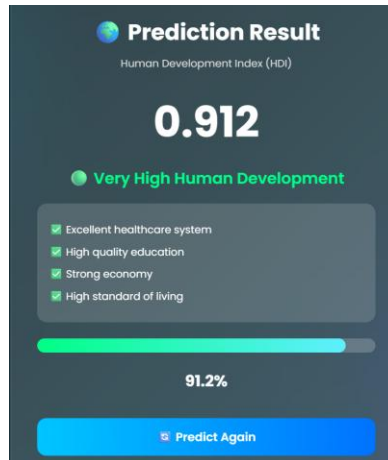
- The Flask application responded quickly to prediction requests.
- The Linear Regression model generated predictions in less than one second for normal inputs.
- No application crashes or runtime exceptions were observed during testing.
- CPU and memory utilization remained well within acceptable limits.
- The application is suitable for educational demonstrations and small-scale usage.

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below:



Screenshot 1: HDI Predictor - Home Page



Screenshot 2: Prediction Result - Very High Human Development